

Adult Income Prediction (Baseline Summary)

1. Problem Framing

Binary classification task: predict whether an individual's annual income exceeds \$50,000 ($1 = >\$50K$, $0 = \leq \$50K$). Business objective: identify individuals likely to earn more than \$50,000 so that marketing and premium-product outreach can be targeted efficiently, reducing wasted contact costs. The target class is imbalanced 23.93% positive ($>\$50K$) vs. 76.07% negative ($\leq \$50K$) which means accuracy alone would be a misleading measure of model quality.

2. Chosen Metric: Precision@top-k

Because outreach budget is limited in practice, the business only ever contacts a fixed-size group of top candidates rather than everyone above a generic threshold. Precision@top-k measures accuracy within exactly that selected group: of the k people flagged as likely high-income, what fraction truly are? This directly reflects the cost of a false positive (a wasted, low-value contact) and extends naturally once a scored model is introduced predictions get ranked by confidence and the top-k highest-confidence candidates are evaluated for precision.

3. Baseline Results (evaluated on the 20% hold-out test set, n = 9,769)

Baseline	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
Majority-class	0.7607	0.0000	0.0000	0.0000	0.5000	0.2393
Rule-based (capital-gain > 0)	0.7823	0.6254	0.2250	0.3309	0.5913	0.3262

The majority-class baseline reaches 76% accuracy purely by exploiting class imbalance, but identifies zero high-income individuals (precision/recall/F1 = 0). The rule-based baseline (capital-gain > 0) beats it on every meaningful metric, reaching **62.5% precision** — equivalently, Precision@k = $526/841 = 0.6254$, where k = 841 is the size of the rule's predicted-positive group. This establishes the benchmark any real model must clear: exceed 62.5% precision while substantially improving recall (currently only 0.225) and F1.

4. Error Analysis

False positives (predicted $>\$50K$, actually $\leq \$50K$): individuals with nonzero capital-gain who still fall in the low-income class — capital-gain alone is not a sufficient signal; occupation, education, and hours worked also matter.

False negatives (predicted $\leq \$50K$, actually $>\$50K$): commonly Bachelor's/Master's degree holders in professional or managerial roles (e.g., Tech-support, Exec-managerial) working 40–60 hours/week with zero capital-gain — high income here comes from salary, not investment income, which the rule entirely misses.

Data/feature issues to address next:

- (1) handle missing values in workclass, occupation, and native-country;
- (2) one-hot encode categorical features;
- (3) log-transform skewed variables (capital-gain, capital-loss);
- (4) engineer combined features from education, occupation, and hours worked;
- (5) train stronger models (Logistic Regression, Random Forest, XGBoost) using ranked probability scores;
- (6) tune hyperparameters via cross-validation while keeping the hold-out test set untouched.

Primary metric going forward: Precision@top-k remains the target metric, since the business goal is minimizing wasted outreach under a fixed contact budget. The rule-based baseline's 62.5% precision is the number to beat a useful model should clear it while meaningfully lifting recall and F1, so more true high-income individuals are captured without degrading contact quality.